

April 14, 2026

Editors
Journal of Computer and System Sciences

Dear Editors,

Please consider the enclosed manuscript, “Zero-Error Recovery from Deterministic Partial Views,” for publication in the Journal of Computer and System Sciences.

The manuscript develops a structural theory of exact recovery under deterministic partial information. Admissible coordinate-subset views induce a confusability graph on latent states, so zero-error recovery becomes a coloring problem. The paper gives an exact characterization of the confusability relations realizable on the full labeled tuple space, identifies block composition with strong graph powers and the resulting Shannon-capacity limit, and develops the corresponding one-shot and asymptotic graph-theoretic framework.

The paper also isolates a structural equality route for the induced graph class: transitive confusability yields a collapse to the cluster-graph case, with meet-witnessing and fiber coherence as sufficient conditions. An affine specialization yields a representable matroid whose rank gives tractable upper bounds on confusability and capacity. The appendix then instantiates the realizability criterion across representative programming-language runtimes, databases, and dependency-management systems, showing that the same exact-recovery coordinates discriminate nontrivially across concrete host models. The manuscript connects zero-error information theory, graph structure, structural classification, and computationally tractable upper bounds in a single exact-recovery framework. I believe it may be of interest to readers of JCSS working across graph-theoretic, information-theoretic, and structural complexity questions.

All theorem-level claims cited in the manuscript are machine-checked in Lean 4. An archived Zenodo record containing the formalization, manuscript PDF, and submission source bundle is available at <https://doi.org/10.5281/zenodo.19354075>. The manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing interests.

Thank you for your time and consideration.

Sincerely,

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